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Roll Number: _____

Amrita Vishwa Vidyapeetham

Mid - Term (Missed)Examination, May 2024

Second Semester

Integrated M. Sc (Mathematics & Physics) / B. Sc Honours in Mathematics

22MAT113 Multivariable Calculus

Time : 2 hours

Maximum Marks: 50

Answer all questions

Qn No.		Marks	CO	Blooms Taxonomy level
1	Find and sketch the domain of the following functions. (a) $f(x,y) = \ln(9 - x^2 - y^2)$ (b) $f(x,y) = \sin^{-1}(y - x)$	4	CO1	L2
2	Evaluate the following limits. (a) $\lim_{(x,y)\to(0,0)} \frac{\sqrt{3x-y}-3}{3x-y-9}$ (b) $\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{(xy)}$	4	CO1	L2
3	Given that $w = 2ye^x - \ln z$, where $x = \ln(t^2+1), y = \tan^{-1} t$, $z = e^t$. Find $\frac{dw}{dt}$ as a function of t also find $\frac{dw}{dt}$ at $t = 1$.	4	CO1	L3
4	Let $f(x,y) = \frac{x-y}{x+y}$. Find the direction $\vec{u}$ such that $D_{\vec{u}} f\left(-\frac{1}{2}, \frac{3}{2}\right) = 1$.	4	CO1	L4
5	Find the direction in which the function $f(x,y) = x^2y + e^{xy}\sin y$ increases most rapidly at the point (1,0).	4	CO1	L2
6	Show that the following limits does not exist	6	CO1	L5

	(a) $\lim_{\substack{(x,y)\to(0,0) \\ y\neq x^2}} \left(\frac{-x}{\sqrt{x^2+y^2}}\right)$ (b) $\lim_{(x,y)\to(0,0)} \left(\frac{x-y}{x+y}\right)$			
7	Find the directional derivative of the function, $f(x,y,z) = e^{yz} + \cos xy + \ln xz$, at the point $(1, 0, \frac{1}{2})$ in the direction of $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$.	6	CO1	L4
8	Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ when $u = \ln 2$, $v = 1$ if $z = 5\tan^{-1} x$ and $x = e^u + \ln v$.	6	CO1	L4
9	Find the equations for the tangent plane and normal line to the surface $\cos \pi x - x^2 y + e^{xz} + yz$ at the point $P_0(0, 1, 2)$.	6	CO1	L3
10	Find a parametric equation for the line tangent to the curve of intersection of the surfaces $x^2 - y^2 - z^2 = 11$ and $xy + yz - zx = 18$ at the point $(6, 4, 3)$.	6	CO1	L5

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create